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Efficient energy use in a slaughter and meat processing plant—opportunities for process integration

Anna Fritzson [a,b,*], Thore Berntsson [b]

[a] SIK—The Swedish Institute for Food and Biotechnology, P.O. Box 5401, SE-402 29 Göteborg, Sweden
[b] Heat and Power Technology, Department of Energy and Environment, Chalmers University of Technology, SE-412 96 Göteborg, Sweden

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Abstract

In this paper, process integration methods are used to investigate the potential to decrease the energy usage in the slaughtering and meat processing industry. Above ambient temperatures, heating of water with different target temperatures is a large heat demand in a plant, while at subambient temperatures the refrigeration plant needs almost all of the shaftwork used at the site. Interaction between, on one hand, energy demands above ambient temperature and, on the other, cooling needs below ambient temperature can take place with freezing compressors or heat pumps. By using process integration methods above and below ambient temperatures, potentials for saving both shaftwork and external heat demand in food plants can be identified. A case study at a modern plant illustrates that even though many energy-saving measures have been taken there is still a technical potential for saving 30% of the external heat demand and more than 10% of the shaftwork used in the plant. The economic potential for the savings is dependent on the conditions at the plant.
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1. Introduction

A number of trends in the food industry will make energy-related issues more important in food processing plants in the future. Firstly, increasing sales of ready-made meals increase the degree of processing in the food industry. This leads to larger energy use in the plant, while the energy usage in households is changed and to some degree transferred to the processing industry (Sonesson, Mattson, Nybrant, & Ohlsson, 2004). Secondly, there is also an increasing demand for greater flexibility in the food processing plants. Consumers want a greater range of different products. This, in turn, increases energy use in the plant e.g. through increased warm water usage. These changes in customer behavior, together with raised energy prices, hardened price competition, and policy instruments such as carbon dioxide trading and taxation, should heighten the interest in saving energy in the food industry.

The food processing plants in Sweden used approximately 5.75 TWh fuel and electricity (Statistics Sweden, 2002) out of the approximately 160 TWh used by all industry in 2000 (EUROSTAT, 2005). However, 20% of the total energy use in Swedish society can be attributed to the food supply system (SEPA, 1997). The overall energy use in Swedish food processing plants consists on average of 45% electricity and 55% purchased fuels, not including fuel for transports. Within the food industry, the dairy and cheese and the slaughtering and meat

* Corresponding author. Address: SIK—The Swedish Institute for Food and Biotechnology, P.O. Box 5401, SE-402 29 Göteborg, Sweden.
E-mail addresses: anna.fritzson@sik.se (A. Fritzson), thore.berntsson@chemeng.chalmers.se (T. Berntsson).

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Nomenclature

CAHP	compression/absorption heat pump
COP	coefficient of performance
GCC	grand composite curve
EGCC	exergy grand composite curve
SMP	slaughter and meat processing
T	temperature [K]
t	temperature [°C]
T_a	ambient temperature, 293.15 K
ΔT_{min}	smallest accepted temperature difference
UC	utility curve
Q	heat/cooling load [kW]

processing industries are particularly electricity-demanding. The sugar industry and drink production plants are particularly fuel-demanding.

Process integration is the common term for system-oriented methods that can be used when designing and retrofitting industrial processes to obtain a complete plant with optimal use of resources, such as energy, raw materials and process equipment. The focus of the methods has traditionally been on efficient energy use, but recently other parameters have been studied.

There are several scientific papers describing case studies using heat pinch analysis in the food industry. Most of them, however, describe industries that are similar to the chemical process industry where the pinch technology was developed such as sugar refineries, dairies and breweries. Relatively few studies have been published in the part of the food industry studied in this paper, represented by the meat industry, where the production is mainly batch-wise and related to the manufacturing industry rather than processing industry. One of these studies was made by Chadderton (1995) describing a pinch analysis in a meat plant in New Zealand. Another study by Fritzson and Vamling (2004) describes an energy analysis using pinch tools in a Swedish slaughter and meat processing plant. The experience in this type of food industry in the consulting sector includes studies in slaughterhouses and ready-made meal plants. Most of these are not available to the public.

The aim of this study is to find the potential to decrease energy use, both external heat demand and electricity use in the refrigeration system, in a modern food processing plant. The Swedish slaughterhouse and meat processing plant (SMP) studied is especially interesting since a modern extensive heat recovery system has been integrated with the plant. In the study, two process integration methods are used to find the potential for saving electricity and fuel demand in the plant. To identify fuel savings, heat pinch analysis is used. To lower the electricity use in the refrigeration plant in the SMP, a shaftwork-targeting methodology based on pinch analysis methodology is used together with process simulation in HYSYS®, an integrated simulation program for continuous processing industries.

2. Method

2.1. Process integration – heat pinch analysis

For the benefit of engineers inexperienced with process integration methods, these techniques are briefly explained in this article. The more practiced engineer is advised to skip this section and proceed to the case study.

Heat pinch analysis is the single most important process integration concept and the one that originally gave birth to the field (Gundersen, 2002). The methodology is based on thermodynamic principles. Using pinch analysis, it is possible to identify appropriate changes in the core process conditions that can have an impact on energy savings. Some questions that can be answered by heat pinch analysis are: What are the minimal external heating and cooling demands for a particular process? What is the maximum heat that can be recovered through internal heat exchangers? How should a heat exchanger network be designed so that internal heat-exchanging is optimized?

The concept of heat pinch analysis was developed into an industrial technology by Bodo Linnhoff and his group at UMIST in Manchester in the 1980s (Linnhoff et al., 1994). To read more about the most basic aspects of process integration methods as well as some of the recent and advanced elements of these methods, see Gundersen (2002).

A heat pinch analysis starts with the heat and material balances for a process. In the analysis, streams that need to be heated (cold streams) or cooled (hot streams) are identified. With these in mind, graphical representations of the energy flows of a process can be produced. Some examples of these graphs are the composite curves (CC) and the grand composite curve (GCC). The GCC shows the energy demands at various process temperatures, the net external heating and cooling demands, as well as the pinch temperature. In this way, targets for energy saving can be set prior to the design of the heat exchanger network.

The pinch divides the process into two separate parts, one part above, and one part below the pinch temperature. In the part that is above the pinch temperature

there is a net heat deficit and external heating is needed. In the part below the pinch temperature there is a net surplus of heat that needs to be removed by cooling the streams. The existing steam consumption is compared with the minimal steam consumption for that process; the difference between them is the potential saving.

The GCC can be used to study how a heat pump or a combined heat and power plant can be integrated with the rest of the process in such a way that the total external energy demand can be reduced. A heat pump should be integrated into the process so that the net surplus of heat below the pinch temperature is used in the evaporator in the heat pump, and so that the heat deficit above the pinch temperature is covered by the excess heat in the condenser.

Since energy savings in one part of a plant are counterproductive if they entail an equal or larger energy increase in another part of the plant, it is important to include the whole site in a process integration study.

2.2. Process integration studies in the food industry

When performing process integration studies, i.e. using heat pinch analysis, in a food processing plant there are several difficulties that need to be overcome. For example, the production volumes and mixes vary with time and demand, there is often no submetering of process consumption, and consequently limited data are available. Additionally, the operating hours per year are fairly few.

Typical for food plants is batch-wise production, which complicates energy recovery. It can be difficult to use all heat sources in a food plant for heat recovery as they are usually not continuous and only available during production times. Contrary to the heat sources, the heat sinks at the site usually occur when there is little or no production, e.g. warm water for cleaning.

Limited data availability complicates the analysis of the time when heat sources and demands are present. One can study the gross potential for saving energy in the plant by starting with a feasibility study of a plant, including the data available and making an average over the yearly production time. The feasibility study will often overestimate the potential for saving external heat and cooling demand, since the heat sources and sinks are seemingly present at the same time. If there is no potential for energy saving, there is no reason to spend time finding more data or making a more thorough analysis of the plant. If there is a large enough potential found in the feasibility study, there is an incentive to make a more detailed study. In the detailed study a more extensive data collection is carried out, including detailed information about when different heat demands and sources occur. Methods for process integration in batch-wise production and energy storage might also be needed. For a review of some of these methods and techniques see Mikkelsen (1998).

2.3. Shaftwork targeting in subambient systems

Traditionally, heat pinch analysis is used to establish heat load targets for the refrigeration levels based on basic process needs without designing the heat exchanger network. Linnhoff and Dhole (1992) describe a way to combine exergy concepts with the principles of pinch analysis, in such a way that targets for the minimal amount of shaftwork needed for a certain refrigeration plant can be identified directly from basic process data. Linnhoff and Dhole propose a curve which is constructed by replacing the temperature on the y-axis with the Carnot factor, $\eta_C = 1T_a/T$, where T_a is the ambient temperature, resulting in the exergy grand composite curve (EGCC).

When using the targeting procedure, an EGCC is constructed for the freezers and refrigeration rooms in the studied plant. The required temperatures in these rooms are reduced by half of the smallest temperature difference (ΔT_{min}) that can be accepted between the refrigeration medium and the required temperature of the air in the freezers and refrigeration rooms. In rooms where the refrigeration medium used is not permitted, additional temperature difference is added to allow for intermediate heat exchange such as a brine heat exchanger. The Carnot factors are calculated for the air temperatures and plotted against the cooling loads. A utility curve (UC) describing the Carnot factors, increased by half of $\Delta T_{min,}$ versus heat load for the temperature levels that are currently used in the refrigeration system is also constructed. For an example of an EGCC and a UC, see Fig. 3. The area between the UC and the EGCC is directly proportional to the excess shaftwork required in the refrigeration plant because of excessively large temperature differences in heat exchangers in the freezers and refrigeration rooms. This means that a decrease in the area between the curves generates a decrease in the shaftwork requirement for the refrigeration plant. In other words, the least possible shaftwork for the refrigeration can be achieved when the EGCC and the UC are perfectly aligned. This, however, usually leads to a refrigeration plant with too many refrigeration levels and therefore a system which is too complex in reality.

Assuming no changes in the refrigeration needs in the food processing plant, the area between the curves can be changed by changing the load for the existing temperature levels in the refrigeration system. This is done so that the cooling needs are achieved with a decreased temperature difference. This gives a decrease in driving force in the coolers, and therefore larger heat exchanger areas are needed. Another way of decreasing the area

between the EGCC of the process and the UC is by increasing the temperature of one or more of the temperature levels in the refrigeration plant to fit the process needs more closely. This leads to an increased capital cost through increased heat exchanger areas and costs for changes in compressors etc. to enable the change in temperature.

The described targeting method enables the designer of a refrigeration system to identify which changes in the refrigeration system are interesting to study more closely without being forced to model every possible solution. It also gives a picture of the whole refrigeration system enabling the designer to see the consequences of changes in rooms with cooling needs for the whole system. Thus, the time needed to find solutions to decrease the shaftwork usage in a refrigeration plant is significantly reduced, and the probability of finding a design that is good for the whole refrigeration system is increased.

2.4. HYSYS®

HYSYS® is a graphic-oriented simulator, developed by Hyprotech Limited with the purpose of modeling, simulating and optimizing continuous chemical plants. In this study, version 3.1 is used. In a HYSYS® model, blocks representing heat exchangers, compressors etc. can be used to simulate refrigeration plants. The temperatures and loads of the refrigeration levels are defined to be similar to those of the plant being modeled and the adiabatic efficiency for the compressors is assumed to be 75% in this study. HYSYS® has previously been used to design an industrial refrigeration plant; see Pang (1998). One part of the refrigeration plant simulated in HYSYS® in this study can be seen in Fig. 1.

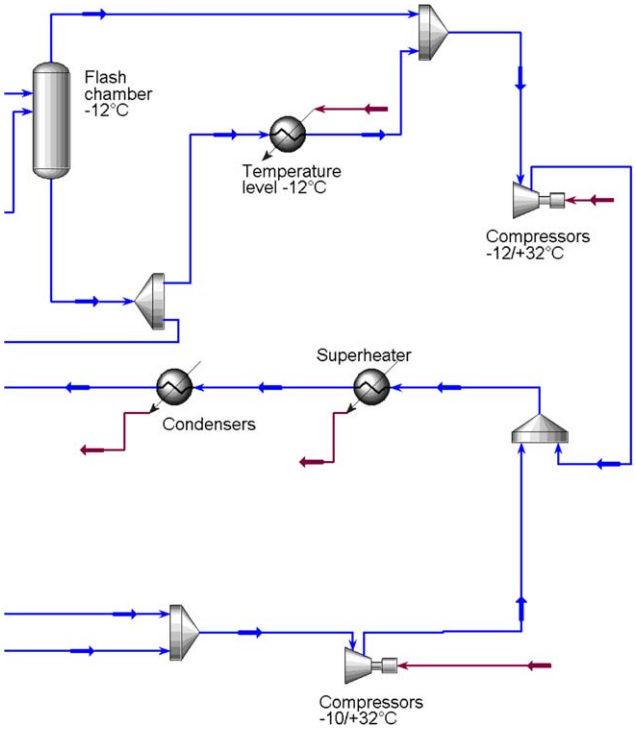


Fig. 1. Part of the refrigeration plant simulated in HYSYS®.

In these simulations, a modified Peng–Robinson equation of state (PRSV) is used for the physical property calculations within the fluid package in HYSYS® (Peng & Robinson, 1976). PRSV is a good model for moderately non-ideal systems. The thermodynamic results from the model were verified in REFPROP which is based on the most accurate pure fluid and mixture models currently available (Lemmon, McLinden, & Huber, 2002). The verification showed a good correlation between the thermodynamic data in the HYSYS® model and in REFPROP.

3. Case study

In this study, a Swedish slaughtering and meat processing plant is examined. Most Swedish slaughterhouses also have some type of meat processing. Plants can differ greatly from each other depending on their production. Earlier, a Swedish plant with a somewhat similar production and energy system has been studied (Fritzson & Vamling, 2004). The studied plant operates approximately 260 days per annum. The products include meat from pork, beef and lamb, ready-made meals and semi-manufactured products. Live animals are transported to the plant where they are slaughtered, dressed and packaged. A large amount of the meat produced in the plant is used in the meat processing plant, where products are prepared for frying and other unit operations. In the plant, an increased production of semi-manufactured products and a higher flexibility demand has been observed during the last decade. The slaughterhouse and meat processing are two facilities with fairly different characteristics from both a process and an energy standpoint.

The energy use in an SMP plant can be divided into two parts: the refrigeration system, which uses electricity to supply energy at subambient temperatures, and streams mostly above ambient, such as water that needs to be heated for use as tap water.

The plant is fairly modern and has an extensive heat recovery system. However, savings can be made by a decrease in warm water usage, decrease in freezing time or more efficient equipment. These options are not included in this study, even though they are currently being studied by staff at the plant. The heat demands in the plant are hot, warm and tap water used primarily for cleaning and hygienic purposes, heating of buildings and processing steam. The cooling demands are cooling

of compressors in the refrigeration plant and compressed air compressors. There is also excess heat available in the singeing flue gases in the slaughterhouse, and it is currently used for heating some of the water used for comfort heating. The plant has two installed heat pumps, and excess heat from these is also used in the plant's heat recovery system.

The excess heat from the refrigeration plant that is not used in the heat recovery system is currently condensed with air and water. The amount of energy in these streams is rather large and is currently not used. This excess energy is especially large during the summer months when the need for comfort heating in the plant is limited.

There is limited energy-data availability in the plant, and therefore the analysis starts with a feasibility study. Average data from the plant's control system and manual measurements, as well as some calculated data, have been used in the analysis of the plant. All the data included in the analysis are checked in a model of the utility system in the plant, in such a way that the data are realistic for an average 24-h period with production in the plant. The data from the plant is confidential and can therefore only be published in graphical form.

3.1. The refrigeration system

The plant currently has two two-stage ammonia refrigeration plants with four refrigeration levels (−10 °C, −12 °C, −25 °C and −40 °C). In the areas of the plant where ammonia is not permitted, brine heat exchangers are used.

3.2. Heat recovery

Excess heat from the compressed air compressors and the compressors in the refrigeration plant is recovered by heat exchange with incoming tap water. The excess heat from the singeing gases is used to heat water for comfort heating. In the refrigeration plant, excess heat from superheated ammonia and condensers is also used to heat tap and warm water. There are also two ammonia heat pumps installed in the refrigeration plant that use some of the excess heat from the refrigeration system to heat hot and warm water and to heat the office buildings in the plant. More than 1 MW internal heat-exchanging is currently used. The heat pumps supply almost 1 MW.

The heating needs that are not covered by heat recovery are supplied by steam. For a layout of the recovery system of the plant, see Fig. 2.

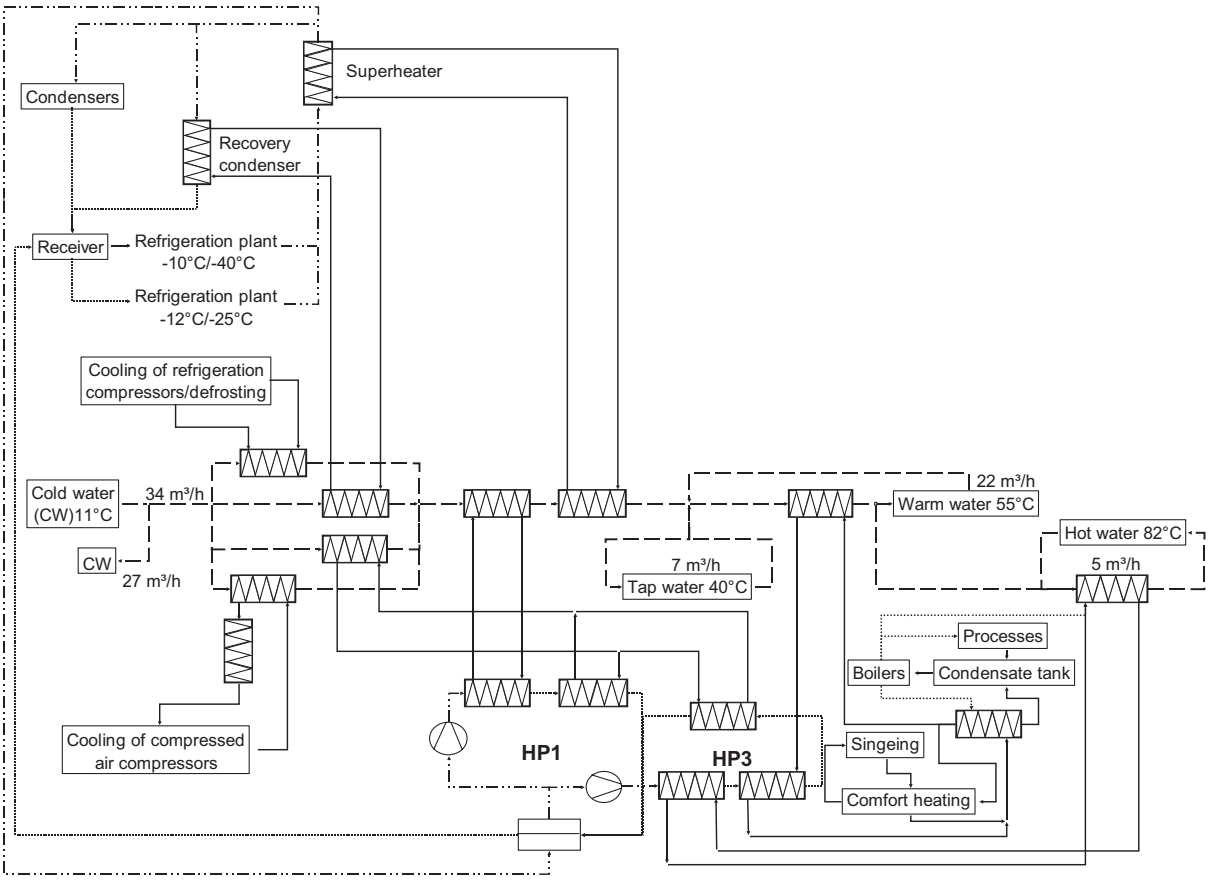


Fig. 2. The heat recovery system in the studied SMP.

4. Results and discussion

4.1. The refrigeration system

The EGCC for the refrigeration needs in the SMP plant are constructed together with the UC describing the current temperature levels and loads supplied by the refrigeration plant. As can be seen in Fig. 3, there is a rather large area between the EGCC and the utility curve indicating a potential for reducing shaftwork in the plant. As a matter of fact, the current shaftwork need is 15% larger than the least possible shaftwork realized when the UC and the EGCC are perfectly aligned. Some of this potential can be fulfilled by using the temperature levels in the refrigeration plant in a more energy-efficient way, and thereby reducing the area between the EGCC and the UC. A smaller area can be accomplished by starting with the load of the temperature level with the highest temperature and adjusting it so that it covers as great a part of the EGCC as possible. All the loads of the temperature levels, from the highest to the lowest, are changed until all have been adjusted to the EGCC; see Fig. 4. In this way, as large a part of the total load as possible is placed on a high temperature

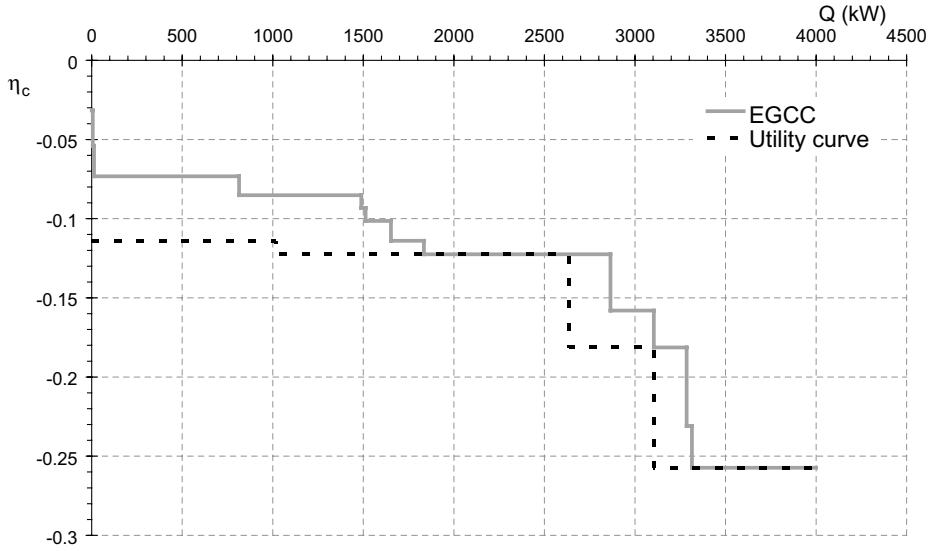


Fig. 3. The EGCC for the refrigeration system needs (shown in grey) and the utility curve describing the current temperatures and loads used in the refrigeration plant. A large area can be seen between these curves indicating a potential to save shaftwork in the refrigeration plant.

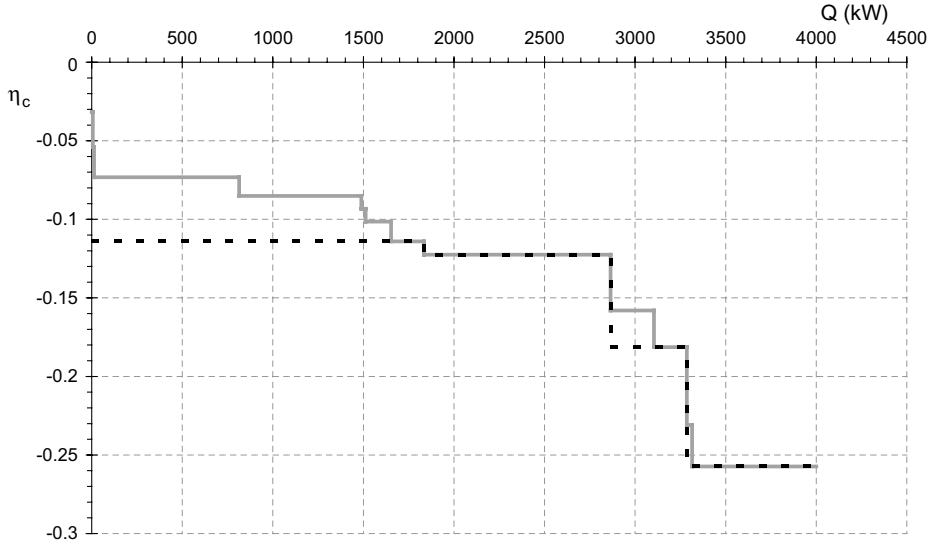


Fig. 4. In order to save shaftwork in the refrigeration plant the area between the EGCC and the utility curve is reduced by fitting the loads of the utility curve to the EGCC.

level, and therefore shaftwork is saved in the refrigeration plant.

Modeling these changes in HYSYS® shows a potential for a shaftwork saving of approximately 5%. This reduction can be accomplished in the plant by increasing the areas of some heat exchangers in the freezers and refrigeration rooms. After these changes the shaftwork need is reduced to a level exceeding the least possible shaftwork by 10%. The air temperature in the freezing and cooling rooms is not changed by these changes.

As can be seen in Fig. 4, after the measures there are still rather large areas left between the EGCC and the UC at the highest temperatures. In order to decrease this area and save additional shaftwork, the temperature of the highest temperature level in the refrigeration plant must be changed to better fit the EGCC. A test of several different temperatures for the highest temperature level shows that if the highest temperature is set to −3 °C and the loads of the temperature levels are adjusted, the least area, and therefore shaftwork, is achieved. The result of this change is shown in Fig. 5. Modeling of this case in HYSYS® shows a potential for a shaftwork saving of more than 10% compared to the current refrigeration plant, not cumulative with the 5% saving. This reduction in shaftwork can be accomplished by necessary changes in compressors and other equipment to enable the change in temperature level and by increasing heat exchanger areas in refrigeration rooms. After these changes the shaftwork need is reduced to only 3% larger than the least possible shaftwork.

The changes in the refrigeration plant suggested above that enable a 5–10% decrease in shaftwork might not be profitable as a stand-alone measure in an existing plant. However, as part of a process change or a renovation project it is probably cost-effective.

It might be interesting to change more than one temperature level in a refrigeration plant to save additional shaftwork. In this case, however, the difference between the used shaftwork and the least possible shaftwork after changing the temperature of the highest level is small, and can be further reduced only by introducing additional temperature levels in the plant. This is probably not profitable in the studied plant, but could be an option when designing a new plant with the same refrigeration needs.

If changes in the room temperatures are possible, an even larger shaftwork saving can be achieved. Another way of saving electricity in the refrigeration plant is by reducing the load in the freezers and refrigeration rooms. For example, if an automatic on/off system based on production times is installed in the freezers that today are manually turned on and off, some electricity can probably be saved. It is also possible to save shaftwork by lowering the condensation temperature. This however, might not be possible if the condensation energy is used somewhere else in the plant.

The studied SMP has a refrigeration plant that is fairly well designed for its refrigeration needs, especially at the −12 °C level where the EGCC and the UC match perfectly. A plant with the same refrigerating needs, but with larger temperature differences between its freezers and refrigerating rooms and its utility system, can achieve larger shaftwork savings by minimizing the area than in the studied SMP. For comparison, a reference case plant with a refrigeration system that is badly designed for its refrigeration needs is made. The reference case has the same EGCC as the studied SMP plant, but

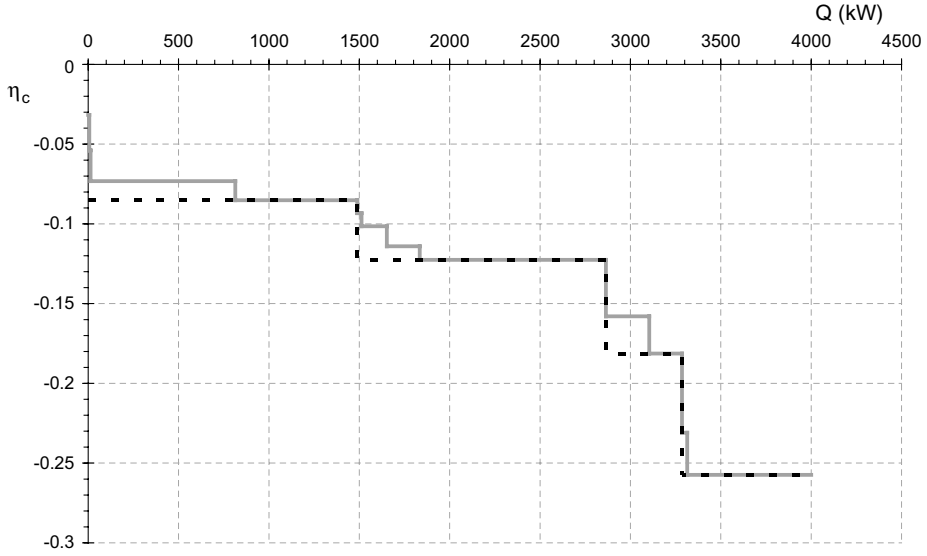


Fig. 5. In order to save shaftwork in the refrigeration plant the area between the EGCC and the utility curve is decreased by changing the temperature of the first refrigeration level from −10 °C to −3 °C and changing the load of all the refrigeration levels to fit the EGCC.

the UC is constructed so that the area between the UC and the EGCC is approximately the same at all temperature levels. In other words, there is a rather large area between the EGCC and the UC representing a shaftwork need 23% larger than the least possible shaftwork in the plant. This is done by changing the second temperature level from −12 °C to −18 °C; see the dotted and dashed line in Fig. 6. According to HYSYS®, when the UC is fitted better to the EGCC by adjusting the loads of the temperature levels in the UC, almost 7% can be saved in the reference case (15% larger than the least possible shaftwork). If two temperature levels are adjusted (−10 °C to −3 °C and −18 °C to −12 °C), almost 16% can be saved compared to the original reference case (3% larger that the least possible shaftwork); see the dashed line in Fig. 6. The changes that are needed to save this amount of shaftwork in the plant are probably large enough to be profitable when there are other reasons for retrofitting the plant.

4.2. Internal heat exchanging

All streams that have temperatures above ambient temperatures are included in the analysis of the possibilities for extended internal heat exchanging. A GCC is constructed for the SMP excluding excess heat from the heat pumps but including the heating needs in terms of water, process steam and comfort heating. The GCC is constructed so that the external heat and cooling demand in the curve is the same as in the plant (see Fig. 7); this is achieved when ΔT_{min} is approximately 7 °C. The large excess heat that can be seen below the

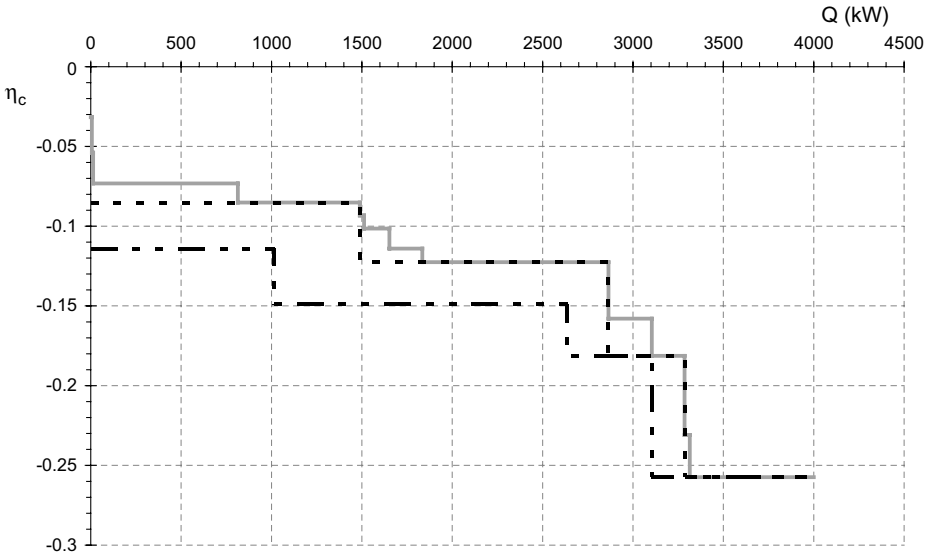


Fig. 6. The EGCC for the reference case. The UC for the original case (−10 °C, −18 °C, −25 °C, −40 °C) is dotted and dashed and the UC for changed loads and temperatures (−3 °C, −12 °C, −25 °C, −40 °C) is dashed.

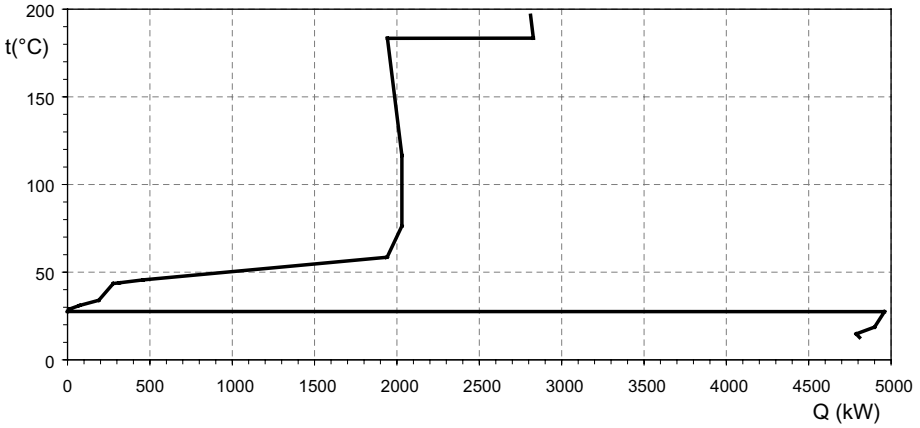


Fig. 7. The GCC for the SMP plant without heat pumps.

pinch is heat from the condensers in the refrigeration plant. The rather low ΔT_{min} indicates that there is not a very large potential for saving external heat and cooling demand by increasing internal heat demand in the plant. If countercurrent liquid/liquid heat exchangers are used, a good industrial practice is to set the temperature difference in the heat exchangers no lower than 5 °C. There is a potential for saving approximately 70 kW in the SMP if the internal heat exchanging is increased so that ΔT_{min} is reduced from 7 °C to 5 °C. This potential would be rather expensive to achieve in the plant, since several more units are needed to improve the heat recovery. This probably also holds true if more exact data is used in the analysis, and there is no reason to make a more detailed study of the plant.

In a plant similar to the SMP, with low heat recovery, there is of course a greater potential for saving external heat and cooling demand. In that case, the measures taken in the SMP are good first steps in reducing energy demands. These are: heat recovery from cooling of compressed air compressors and refrigeration compressors, heat recovery from the refrigeration plants in condensers and superheaters, and heat recovery from the singeing flue gases.

When designing an extensive heat recovery system, it is important to investigate the heat excess and demands in the plant to find out how their loads vary over time. As mentioned earlier, some of the sources are present during the daytime and some of the demands are present during the evening. It is, however, possible to design a profitable heat recovery system even though these variations exist, as can be seen in the SMP.

4.3. Interactions between subambient and above-ambient systems

After constructing the GCC, it is possible to study the potential for interaction between the streams mainly above ambient temperatures and the heat excess from the refrigeration plant. This interaction can take place using a heat pump. The excess heat from the heat pumps that are already installed in the SMP can be illustrated in the GCC; see the grey curve in Fig. 8. As can be seen, there is a rather large part of the heat demand that can be covered by installing a new heat pump, approximately 1 MW. The suggested condenser temperature of this new heat pump is 62 °C. This is at a rather high pressure, 27 bar, for a traditional ammonia system but it is feasible with modern-type compressors.

Instead of using ammonia as a refrigerant, R134a can be used. A heat pump with R134a has a COP of approximately 6, and a lower investment cost since the pressure needed is lower, 17.6 bar. Another option, if ammonia is the working fluid of choice, is a single-stage compression/absorption heat pump (CAHP) with an ammonia/water mixture as the working fluid. Hultén and Berntsson (2002) show in their study that an industrial single-stage CAHP using an ammonia/water mixture can be used at the required temperatures and yield a good COP. In the study of a case where the condenser temperature is 65–70 °C and the source temperature 35–40 °C, a COP of 5.3 is reached.

In the SMP, almost 30% of the heat demand in the plant is covered by the internal heat exchanging already installed. The installed heat pumps contribute 1 MW, and therefore less than 50% of the heat demand is currently covered by steam from the boilers. If the steam needed in the process, for example for heating in the meat-processing part of the plant, is excluded, 65% of the heat demand is covered by internal heat exchanging and the heat pumps. If another heat pump is installed in the plant, as suggested above, almost all of the heat demand, excluding process steam, can be supplied by internal heat exchanging and heat pumps.

To find out if the installation of a new heat pump is profitable for the SMP, more detailed data are needed

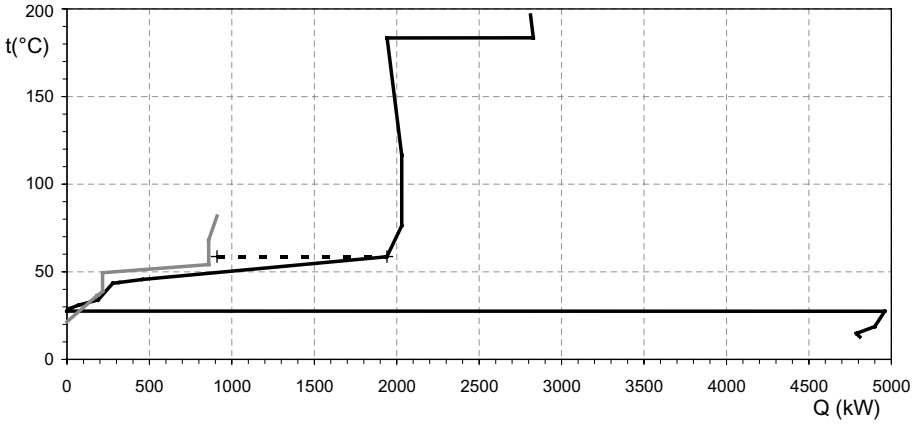


Fig. 8. The GCC for the SMP with the grey curve showing the temperature versus heat load for the installed heat pumps. The dashed line shows the opportunity to install an extra heat pump using the excess heat from the condensers in the refrigeration plant.

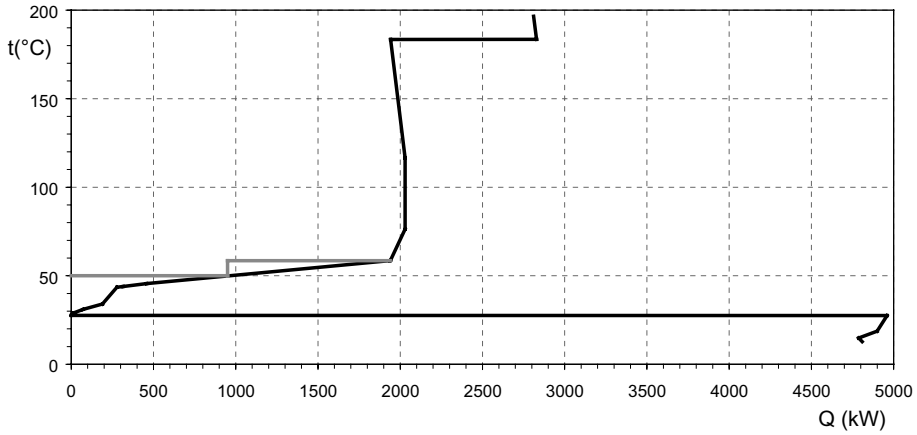


Fig. 9. Two new heat pumps can be integrated with the existing SMP or a plant with a similar GCC.

to reveal the variation in energy demand and excess during different seasons, weeks and times of the day. However, this study shows that there is a large enough potential for saving energy by installing a new heat pump to motivate further studies. It is probable that the size of the heat pump needs to be larger than shown by the GCC constructed with average data from the plant, in order to deliver the load needed during peak hours if there is no possibility of thermal storage. Additionally, it is likely that the pump is not needed during the summertime when no heating of buildings is needed. The study also shows that the excess from the refrigeration plant is large enough to install additional heat pumps, even if the shaftwork needed in the plant is decreased according to the suggestions made in this study.

If a new plant is to be built with a GCC similar to the existing SMP, it is, according to the GCC, justified to install two heat pumps with larger heat loads than the existing ones, thus better satisfying the heat needs in the process; see Fig. 9. By doing this, fuel is only needed for the production of processing steam in the SMP plant.

4.4. Economy

A 10% saving of shaftwork in the refrigeration system, by changing some loads and one temperature level in the refrigeration plants, yields a decrease of €40 500 [1] per annum at typical Swedish electricity prices in 2004 (€35 per MWh) and a possibility to save even more as the price of electricity will probably increase in the future.

The new suggested heat pump should not be less profitable than the two that are already installed in the plant. The investment cost of the high-pressure heat pump might be larger, but the investment cost for a heat pump using R134a might even be 20–25% lower than for a traditional ammonia pump (Ingvarsson, Steén Ronnermark, & Eriksson, 2004).

More exact data are needed to make a good economic evaluation of the results in this study. It is, however, probable that the profitability of the suggested refrigeration changes and installation of a heat pump is dependent on the plant being a new design, or being given a retrofit for other reasons.

5. Conclusions

In the modern plant studied, there is a low potential for extended heat recovery by internal heat exchanging in the plant. However, a less modern plant can, by using the same heat recovery methods as the studied SMP, save 30% of the external heat demand fairly easily.

A potential to save electricity by redesigning the refrigeration plant has been found. More than 10% of the shaftwork used in the refrigeration plant can be saved in the modern SMP plant and 16% can be saved in a less modern plant.

The largest potential found in this study can be realized by installing an additional heat pump. There is excess heat available in the refrigeration plant and a potential for saving almost all of the external fuel demand, excluding processing steam, by installing a new heat pump. In a plant without prior heat pumps, the installation of two heat pumps can also save the same amount. Some of the proposed changes are probably profitable to make in a new design or during a retrofit of the plant.

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[1] €1 = 9.13 SEK. This corresponds to the average exchange rate between March 2002 and March 2005 (European Central Bank, 2005).

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References

Chadderton, T. (1995). Using pinch analysis to identify opportunities to reduce opportunities to reduce meat plant utility costs. *MIRINZ 940*. Meat Industry Research Institute of New Zealand, Hamilton, New Zealand.

European Central Bank, (2005). http://www.ecb.int/stats/exchange/eurofxref/html/eurofxref-graph-sek.en.html, accessed March 20, 2005.

EUROSTAT, (2005). http://epp.eurostat.cec.eu.int, Final energy consumption by industry, accessed May 5, 2005.

Fritzson, A., & Vamling, L. (2004). Energianalys vid slakterier—en förstudie (Energy analysis at slaughterhouses—a feasibility study). In Swedish. *PRO-04/3*. The Swedish Energy Agency, Eskilstuna, Sweden.

Gundersen, T. (2002). IEA: A Process Integration Primer.

Hultén, M., & Berntsson, T. (2002). The compression/absorption heat pump cycle—Conceptual design improvements and comparisons with the compression cycle. *International Journal of Refrigeration, 25*(4), 487–497.

Ingvarsson, P., Steén Ronnermark, I., & Eriksson, M. (2004). Naturliga köldmedier—framtida värmepumpar för fjärrvärme (Natural refrigerants—future heat pumps for district heating). In Swedish. *A4-303*. Värmeforsk, Stockholm.

Lemmon, E. W., McLinden, M. O., & Huber, M. L. (2002). REFPROP Reference fluid thermodynamic and transport properties.

Linnhoff, B., & Dhole, V. R. (1992). Shaftwork targets for low-temperature process design. *Chemical Engineering Science, 47*(8), 2081–2091.

Linnhoff, B., Townsend, D. W., Boland, D., Hewitt, G. F., Thomas, B. E. A., Guy, A. R., & Marsland, R. H. (1994). A user guide on process integration for the efficient use of energy. IChemE, Rugby, UK.

Mikkelsen, J. B. (1998). Thermal-energy storage systems in batch processing. Ph.D. thesis, Department of Energy Engineering, Technical University of Denmark, Copenhagen.

Pang, K. H. (1998). A novel use of HYSYS® to design an industrial refrigeration system. Session 3613. In *Proceedings of the 1998 Annual ASEE Conference, Seattle, WA, USA*.

Peng, D. Y., & Robinson, D. B. (1976). A new two constant equation of state. *Industrial and Engineering Chemistry, Fundamentals, 15*(1), 59–64.

SEPA, (1997). Att äta för en bättre miljö (To eat for a better environment). In Swedish. *Report 4830*. Swedish Environmental Protection Agency, Stockholm, Sweden.

Sonesson, U., Mattson, B., Nybrant, T., & Ohlsson, T. (2004). Industrial processing versus home cooking—an environmental comparison between three ways to prepare a meal. *Ambio, 34*(4–5), 411–418.

Statistics Sweden, (2002). Compiled data. Stockholm, Sweden.

Anna Fritzson is a Ph.D. student working at SIK—the Swedish Institute for Food and Biotechnology and at Heat and Power Technology, Department of Energy and Environment, Chalmers University of Technology.

Thore Berntsson is the professor of Heat and Power Technology, Department of Energy and Environment, Chalmers University of Technology.